

# Ashish Molakalapalli

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## EDUCATION

### Master of Science in Computer Science

Texas A&M University

Aug 2025 – May 2027

College Station, TX

- GPA: 3.833 / 4.0 | Graduating **May 2027** | Coursework: Analysis of Algorithms, Operating Systems, Machine Learning, Artificial Intelligence, NLP Foundations & Tech, Information Storage & Retrieval
- Focus: Data Structures & Algorithms, Distributed Systems, Scalable Backend Services, API Design, Cloud Infrastructure

### Bachelor of Technology in Computer Science & Engineering

IIIT Kottayam

Dec 2021 – Apr 2025

India

- GPA: 9.34 / 10.0 | Selected Research Scholar, IoT Cloud Research Group – Top 5 of 150+ applicants

## INDUSTRY EXPERIENCE

### Software Engineering Intern – Security Platform

ServiceNow

May 2025 – Aug 2025

San Diego, CA

- Designed and shipped a modular 14-tool service layer in JavaScript orchestrating parallel database queries, REST callouts, and scheduled-job introspection across production security infrastructure with zero manual intervention.
- Built a schema-driven analysis pipeline with a 14-field JSON contract and confidence-based escalation logic; integrated AWS Bedrock as the inference backend, achieving consistent structured output across all instance types in production.
- Engineered a real-time alerting service delivering health summaries to Microsoft Teams via webhooks, replacing a manual multi-instance review process and cutting incident response latency from hours to minutes.
- Resolved cross-environment deployment conflicts during a production migration by auditing update sets, isolating conflicting records, and establishing a repeatable release protocol adopted for subsequent releases.

### Software Engineering Intern

Shipsy

Sep 2024 – Jul 2025

Gurugram, India

- Engineered event-driven backend automation integrating Slack, Gmail, and GitHub REST APIs to process 10K+ requests per day for DTDC; implemented request batching and async processing, reducing manual triage by 35% and cutting average API latency by 40%.
- Built a prototype knowledge graph service with structured entity linking and full-text search, improving internal documentation query response time by 40% under concurrent load.
- Designed and shipped an autonomous standup transcription pipeline with retry logic, idempotency guarantees, and Jira write-back; won Best Idea Award at Shipsy Hackathon 2025.

### Software Developer Intern

UST Global

Jun 2024 – Aug 2024

Kochi, India

- Architected SmartChat as a modular microservices system; exposed NLP inference pipelines via production-grade REST endpoints with structured request validation, error handling, and response observability.
- Optimized request routing and introduced query-time caching, reducing average resolution latency by 30% and improving end-to-end accuracy by 25% under sustained load.

### Software Intern

Nav Tech

Dec 2023 – Feb 2024

Hyderabad, India

- Implemented low-latency ingestion APIs for hardware data streams using async I/O and connection pooling, reducing average response time by 45%.
- Introduced health-check endpoints and alerting hooks integrated with the monitoring stack, sustaining 99.9% service uptime across the device fleet.

## PROJECTS

### Real-Time Job Ingestion and Deduplication Platform | Python, GitHub Actions, Concurrent I/O, REST APIs, JavaScript

- Built a concurrent, horizontally scalable ingestion service in Python consuming 950+ external REST APIs across 6 platforms, sweeping 52K+ records per run with a bounded thread pool and per-source fault isolation.
- Designed the deduplication algorithms and data structures collapsing 4K raw records into 3.2K unique entries via two-pass normalized entity matching and canonical URL matching, with a self-growing registry that discovers new sources unattended.

## Scalable Task Management Platform | *React, Node.js, WebSockets, PostgreSQL, AWS EC2/RDS, Docker, JWT*

- Built a full-stack real-time collaboration application with a React front-end and Node.js back-end, WebSocket-driven live updates and optimistic UI; deployed containerized services on AWS EC2 with RDS PostgreSQL, sustaining 99.9% uptime under concurrent load.
- Implemented JWT authentication, role-based access control, and database indexing strategies to handle high-concurrency workloads without lock contention; load-tested to 500 concurrent users with sub-200ms P99 latency.

## Financial Document Intelligence: Production RAG System | *RAG, GPT-4o, FAISS, BM25 Hybrid Search, Python, Flask, Docker, AWS EC2*

- Designed and deployed a production document search service over 100+ financial prospectuses using dense FAISS vector indexing with BM25 sparse re-ranking; improved retrieval precision by 32% over a TF-IDF baseline.
- Containerized the service with Docker and deployed on EC2; exposed low-latency REST endpoints with per-query source attribution, reducing analyst document retrieval time by 50% in production use.

## E-Commerce Order Management Platform | *React, Node.js, MySQL, AWS ALB/RDS, Auto Scaling, Docker*

- Orchestrated checkout and payment-order REST APIs with rate limiting, idempotency keys, and request validation; modeled the relational schema on RDS MySQL ensuring transactional integrity under concurrent load.
- Scaled horizontally on AWS via ALB and auto-scaling groups; sustained sub-300ms response times under 10x traffic spikes without manual intervention.

## TECHNICAL SKILLS

**Languages:** Python, JavaScript (Node.js, React), C++, Java, SQL, Bash

**Backend & Systems:** REST APIs, Microservices, WebSockets, Async I/O, Event-Driven Architecture, Rate Limiting, Message Queuing, Concurrency

**Databases:** PostgreSQL, MySQL, Query Optimization, Indexing, Transactions, FAISS (vector search), Redis

**Cloud & DevOps:** AWS (EC2, RDS, S3, ALB, SageMaker, Bedrock, Auto Scaling), Docker, Git, CI/CD, GitHub Actions, SLURM

**Tools & Platforms:** ServiceNow (Skill Kit, GlideRecord, Update Sets), Flask, n8n, Weights & Biases, Jira, Pandas, NumPy

**CS Fundamentals:** Distributed Systems, System Design, Data Structures & Algorithms, OS Concepts, Networking, Concurrency

## RESEARCH EXPERIENCE

### Research Assistant, NLP Lab

*Feb 2026 – Present*

**Texas A&M University**

*College Station, TX*

- Built an automated data-ingestion and feature-engineering pipeline in Python converting raw cluster telemetry from 941 profiled jobs across 9 GPU configurations into a structured 37-feature dataset used for downstream model training.
- Extended the benchmark with a 1,884-run LLM inference dataset capturing prefill/decode split timing, TTFT, TPOT, and KV-cache pressure across 22 models and 4 GPU types.
- Trained gradient-boosted and neural regressors on the resulting dataset, reaching 23.7% MAPE ( $R^2 = 0.881$ ) on training-job runtime and 8.8% MAPE ( $R^2 = 0.982$ ) on inference latency.
- Instrumented distributed training runs across DistributedDataParallel, DataParallel, and DeepSpeed ZeRO stages 1–3 to capture multi-GPU scaling behaviour under mixed precision.

### Graduate Researcher, NLP Lab

*Sep 2025 – Feb 2026*

**Texas A&M University**

*College Station, TX*

- Designed an 8-stage automated data processing pipeline converting P&ID engineering schematics to structured JSON graph outputs; integrated YOLOv8 for component detection, PaddleOCR with EasyOCR for multi-scale text extraction, and HDBSCAN with UMAP for symbol clustering.
- Containerized the pipeline as a REST API service with structured JSON output contracts, enabling integration with downstream LLM reasoning and knowledge graph construction systems.

## ACHIEVEMENTS & PUBLICATIONS

- **Hackathon Winner** – Best Idea Award, Shipy Hackathon 2025 (40+ teams) for an autonomous standup-to-Jira pipeline; National Finalist, Smart India Hackathon competing against 500K+ participants across institute, regional, and national rounds.
- **AI/ML Club Sub-Lead**, IIIT Kottayam (2024) – Directed 8 ML workshops and 3 hackathons; grew active membership by 40% and mentored 30+ students. **IEEE Event Organizer** (2023) – Hosted the IoT Cloud Research Symposium with 200+ attendees across 6 research sessions.
- **Findings of EMNLP 2026 (Accepted)** – *Deep Learning Job Runtime Prediction for GPU Cluster Scheduling*; co-authored survey on deep learning job runtime prediction for GPU cluster scheduling, in collaboration with XPerf and Texas A&M University, grounded in a 941-job, 37-feature benchmark across 9 GPU configurations plus a 1,884-run inference dataset.
- **CBSE Merit Certificate** – Perfect score 100/100 in Mathematics (Grade 10), awarded to the Top 0.1% of all national candidates.